AFC–10382

Sub. Code
1BCE1C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Computer Science

PROGRAMMING IN C AND DATA STRUCTURES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is identifier?

2. What is the purpose of gets and puts functions?

3. Define array.

4. What is pointer?

5. What is Enum data type?

6. What is the difference between array and structure?

7. Convert the following expression into postfix form. $(a+b)/(c+d)$.

8. What is linked stack?

9. Define degree of a tree.

10. What is depth of a tree?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Compare do.. while and while .. do loop.

Or

(b) What is recursion? Explain with an example.

12. (a) Explain any five string functions with example.

Or

(b) Write a 'C' program to arrange 10 names in alphabet.

13. (a) Explain any five operations related with files.

Or

(b) Write a 'C' program to illustrate Enum.

14. (a) Writ an Algorithm to delete an node from the linked list.

Or

(b) Explain push and pop operations in a stack.

15. (a) Explain representation of Binary tree using array.

Or

(b) How to represent binary tree using linked list?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss Datatypes supported by C.

17. Explain Dynamic memory allocation.

18. Write a 'C' program to declare results of an examination for ten students using structure.

19. Explain Representation and operations on queue.

20. Discuss applications of Binary tree.

AFC–10383

Sub. Code
1BCE2C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Second Semester

Computer Science

PROGRAMMING IN C++ AND ALGORITHMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Class.

2. What is Private member function?

3. What is Constant member function?

4. How do create array of objects?

5. What is Default Constructor?

6. Define Destructor.

7. Define polymorphism.

8. What is pure virtual function?

9. What is the principle used in quicksort?

10. How is Graph represented using Adjacency matrix?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Compare switch statement with if ..... else if statement.

Or

(b) What is inline function? Give an example.

12. (a) Explain passing objects as function argument.

Or

(b) Explain friend function with an example.

13. (a) Explain parametreized constructor with an example.

Or

(b) Explain copy constructor.

14. (a) Write a C++ program to find out area of Triangle, square and rectangle using function over loading and virtual function.

Or

(b) Write a C++ program to illustrate abstruct class.

15. (a) Explain Binary search algorithm with an example.

Or

(b) Explain merge sort with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss principles of object oriented programming.

17. Explain static member function and pointer to member.

18. Explain Dynamic constructor and initialization of objects.

19. Explain multiple and hybrid inheritance.

20. Explain prim's algorithm.

AFC–10384

Sub. Code
1BCE3C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Computer Science

DATABASE MANAGEMENT SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is database system?

2. Write any two E-R design issues.

3. What is normalization?

4. What is functional dependency?

5. Define transactions.

6. State the benefits of distributed transactions.

7. What is Data Integrity?

8. Define View.

9. Write the benefits of Stored procedure.

10. State the uses of Packages.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the semi structured databases in detail.

Or

(b) What is E-R model? Explain its components of it.

12. (a) Write a short note on First Normal form.

Or

(b) Illustrate about the functional dependency theory.

13. (a) Elucidate the Server system architecture in detail.

Or

(b) Briefly discuss about Distributed systems.

14. (a) Explain the procedures to create Indexes with an example.

Or

(b) Illustrate the steps involved in creating Synonyms with an example.

15. (a) How do you create the implicit cursor in SQL? Explain with an example.

Or

(b) Write the features of PL/SQL.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Briefly explain the transaction management.

17. Discuss the 4NF with an example.

18. Give a brief account on parallel databases.

19. Briefly explain the concepts of Data Integrity.

20. Write a PL/SQL program to find the biggest one from given n numbers using functions.

———————

AFC–10385

Sub. Code
1BCE4C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fourth Semester

Computer Science

JAVA PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is JVM?

2. What are the data types of Java?

3. What are the bitwise operators of Java?

4. Find the value of X? if X= 123.56% 10.

5. Define Constructor.

6. What are the advantages of Vectors over arrays?

7. What are the advantages of Multithread?

8. How will you hide a class in a package?

9. How Applets differs from and Applications?

10. Write the syntax and an example of the method which is used to draw a rectangle.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the features of Java.

Or

(b) Explain the Java support systems.

12. (a) Explain any ten mathematical functions in Java with example.

Or

(b) Explain the switch statement in Java with an example.

13. (a) Write a java program to define an interface Area. Implement the interface to calculate the area of a rectangle.

Or

(b) Explain method overloading with an example.

14. (a) Explain the Thread Life cycle.

Or

(b) Explain the run time error in Java with a program.

15. (a) Explain the Applet life cycle.

Or

(b) How will you add an Applet to a HTML file? Explain with an example.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain Command Line Arguments with a program.

17. Write Java programs to print the first 20 prime numbers.

18. Write a Java program to multiply two matrices.

19. Write a Java program to create a package called student. Import the package to calculate the total mark obtained by the student.

20. Write a Java program to draw a polygon with five sides.

AFC–10386

Sub. Code
1BCE5C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Science

OPERATING SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Operating systems.

2. What do you mean by segmentation?

3. List out the different states of processes.

4. Mention the four necessary conditions for deadlock.

5. Bring out the advantages of concurrent programming.

6. How communications can be done among devices?

7. List down the various responsibilities of file manager.

8. Define data compression?

9. What is kernel in Unix?

10. How the files can be organized in UNIX?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write a short note on demand paged memory management.

Or

(b) Describe the advantages of dynamic partitions in memory.

12. (a) Give a note on cache memory.

Or

(b) Define deadlock. How deadlocks can be modeled in a system?

13. (a) Brief a note on process synchronization mechanisms.

Or

(b) Explicate direct access storage devices.

14. (a) How can the file be organized? Discuss it.

Or

(b) Briefly explicate on access control verification module.

15. (a) Illustrate the history of Unix OS.

Or

(b) How can devices be managed in Unix? Describe it.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain about the paged memory allocation in detail.

17. Discuss the various strategies for handling deadlocks.

18. Explicate on components of I/O subsystem.

19. Compare the advantages and disadvantages of Network OS and Distributed OS.

20. Illustrate in detail the processor management technique in Unix.

———————

AFC–10387

Sub. Code
1BCE5C2

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Science

COMPUTER ORGANIZATION AND ARCHITECTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the basic symbols are used for Register transfers?

2. Write the micro operations for logical AND, OR, NOT.

3. Draw the Block diagram for Input, Output configuration.

4. Write the functions of Interrupt cycle.

5. Write the major components of CPU.

6. Write the principle of register stack.

7. Draw the Hardware for signed 2's component addition and subtraction.

8. Draw the block diagram of CPU-IOP communication.

9. What is meant by locality of reference?

10. Write the differences between ROM and RAM.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the functions of computer registers.

Or

(b) Explain about three-state bus buffers.

12. (a) Explain about program loops.

Or

(b) Explain the flow chart for multiplication program.

13. (a) Describe about the types of Addressing modes.

Or

(b) Explain about the RISC computers.

14. (a) Describe about DMA transfer.

Or

(b) Describe with flow chart Booth's Multiplication Algorithm.

15. (a) Explain the page replacement mechanism in virtual memory.

Or

(b) Explain the mechanism of Associative memory.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe about timing and control unit.

17. Explain about memory reference instructions.

18. Explain about stack organization.

19. Explain with flow chart of division algorithm of fixed point numbers.

20. Describe the characteristics of multiprocessor.

AFC–10388

Sub. Code
1BCEE1A

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Science

Elective: DATA MINING AND DATA WAREHOUSING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is delivery process?

2. What is Transforrm data?

3. What are the functions of tools in data warehosue?

4. How is the data load turned?

5. What is Datamining?

6. What are the basics of datamining?

7. What is similarity measures?

8. How are rural networks used in datamining?

9. Define large item set.

10. What is confidence level and support level is association rules?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What is data ware house? Explain.

Or

(b) Describe briefly about functions of load manager.

12. (a) Explain about assessing performance in data warehouse.

Or

(b) Explain the functions of query manager.

13. (a) Explain about data mining metrics.

Or

(b) Write briefly about data mining from a database perspective.

14. (a) How are Fuzzy sets and Fuzy logic used in data mining.

Or

(b) Describe about decision support system.

15. (a) Describe about Incremental rules.

Or

(b) Explain about measuring the quality of rules.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain about extract and load process, Back up and archieve process in data warehouse.

17. Briefly discuss about capacity planning and tuning.

18. Discuss Data mining issues.

19. Explain about statistical perspective on data mining.

20. Explain about parallel algorithm.

AFC–10389

Sub. Code
1BCEE1B

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Science

Elective : DIGITAL IMAGE PROCESSING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Sampling.

2. What is Imaging geometry?

3. Write expression for discrete fourier transform.

4. What is need for fourier transform.

5. What is block circulant matrix?

6. What is image resortation?

7. What is Boundary detection?

8. Define Thresholding in image segmentation.

9. Define open and close operation in morphology.

10. What is Boundary Descriptor?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain elements of visual perception.

Or

(b) Write short notes on photographic films.

12. (a) Explain fast fourier transform.

Or

(b) Explain hotelling transform.

13. (a) Explain Algebraic approach to restoration.

Or

(b) How is geometric transformation used in Image restoration?

14. (a) What is region oriented segmentation? Explain .

Or

(b) Write short notes on Lossy compression.

15. (a) Discuss about regional descriptors.

Or

(b) Explain about relational descriptors.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain about various steps involved in image processing.

17. Explain the properties of two dimensional Fourier transform.

18. Explain Image Enhancement in the frequency domain.

19. Explain Image compression model.

20. Explain steps involved in Image analysis.

———————

AFC–10390

Sub. Code
1BCEE2A

B.Sc. DEGREE EXAMINATION, NOVEMBER2017

Fifth Semester

Computer Science

Elective : WEB DESIGN TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. How the special characters are inserted into a webpage?
2. How will you connect the two webpages?
3. How will you create the drop down menu in a web page?
4. What is meant by Embedded style sheet?
5. How conditional operators are used in Java Script?
6. Write the operator procedure of java script.
7. How functions are used as object in Java Script?
8. Write the syntax of math object.
9. What are the uses of onfocus and onblure event?
10. What are the uses of XML name spaces?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) How horizontal rules, line breaks are created in HTML programming?

Or

(b) Design a calender(monthly calendar) using HTML.

12. (a) How will you link the external style sheet?

Or

(b) How will you set the Backgrounds for a web page?

13. (a) Write the syntax of increment, decrement operatosr with example.

Or

(b) Write the syntax of IF structure with example.

14. (a) Explain about the window object in detail.

Or

(b) Explain the Searching arrays concept in Java Script.

15. (a) Describe about the event handlers.

Or

(b) Explain about the DOM tree.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Design a web page which shows the temples, monuments and some important tourist places in Tamil Nadu.

17. Explain the element dimension in Cascading style sheets.

18. Explain the logical operators with example in Java script.

19. Describe the Booleam and Number object in Javascript.

20. Describe the Basic Principles and structuring of data in XML.

AFC–10392

Sub. Code
1BCE6C1

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Sixth Semester

Computer Science

COMPUTER NETWORKS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the components of a computer network communication?

2. Name the network topologies.

3. What are the cables used in guided media?

4. What are the propagation methods available in wireless transmission?

5. What are the duties of data link layer?

6. What is meant by burst error?

7. What is meant by multicasting?

8. What are the types of routing techniques present in network layer?

9. What are the types of domain available in Internet?

10. What are the functions of Mail Transfer Agent (MTA)?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the categories of Networks.

Or

(b) Explain the network standards.

12. (a) Describe about the fibre optic cable.

Or

(b) Explain about the communication satellites.

13. (a) Describe about sliding window protocol.

Or

(b) Explain about the Ethernet.

14. (a) Explain the multiplexing and demultiplexing process in Transport Layer.

Or

(b) Write the ports used by the transmission control protocol.

15. (a) Explain the components of cryptography.

Or

(b) Describe about traditional ciphers in symmetric key cryptography.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the reference models of the computer network.

17. Describe about the mobile telephone system.

18. Describe about the any two error detection methods.

19. Explain the User Datagram Protocol (UDP) in detail.

20. Explain about the World Wide Web.

———————

AFC–10393

Sub. Code
1BCE6C2

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Computer Science

COMPUTER GRAPHICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. How do find out the distance between a point and a line?

2. What is antialiasing?

3. What is display – file?

4. How do best whether a point is inside a polygon or not?

5. What is the purpose of homogeneous co-ordinates?

6. What is the structure of segment table?

7. Define window.

8. What is clipping?

9. What are interactive input graphics devices?

10. What is echoing?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is frame buffer? Explain its uses.

Or

(b) Write short notes on character generation.

12. (a) What are the functions of display – file interpreter?

Or

(b) Explain different line style primitives.

13. (a) How do create, delete and rename a segment?

Or

(b) Explain about raster technique.

14. (a) Derive matrix for viewing transformation.

Or

(b) Explain Cohen – Sutherland clipping algorithm.

15. (a) Explain about event handling.

Or

(b) Explain about simulating a locator.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain Bresenham's line generation algorithm.

17. Explain polygon filling algorithm.

18. Derive matrix representation for sealing, translation and rotation.

19. Explain Sutherland – Hodgman algorithm.

20. Explain about interactive techniques.

AFC–10394

Sub. Code
1BCE6C3

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Computer Science

SOFTWARE ENGINEERING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the technical parameters of medium size projects.

2. Write the format of a system definition.

3. Write the development effort multiplier for software reliability.

4. Write the format of a software requirements specification.

5. Write the desirable properties of a modular system.

6. Write the guidelines for good coding style.

7. What is meant by integration testing?

8. Write the five laws of program evaluation.

9. What are the needs for the quality of a software product?

10. What are the factors determine the quality of a software product?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain how the programmers spend their time.

Or

(b) Describe the factors to consider in project planning.

12. (a) Explain about Delphi cost estimation.

Or

(b) Describe about the recurrence relations.

13. (a) Explain the types of software system structure.

Or

(b) Explain the disadvantages of GOTO statement.

14. (a) Explain the art of debugging.

Or

(b) Describe about source code metrics.

15. (a) Describe the ISO 9000 quality standard.

Or

(b) Explain the quality concepts in software product.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain about the programming team structure.

17. Explain the algorithmic cost models.

18. Explain about the distributed system design.

19. Explain about the unit testing procedure.

20. Describe about the statistical quality assurance.

AFC–10395

Sub. Code
1BCEE3A

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Computer Science

***Elective* — MOBILE COMMUNICATION**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Draw the frequency spectrum diagram.

2. Write the frequency ranges used for radio transmission.

3. What is meant by cyclic repetition of data?

4. Define : DDMA technology.

5. Which principle is used in wireless LAN?

6. What are the functions of radio access layer?

7. What is meant by mobile IP?

8. What is meant by time out freezing?

9. What are the functions of wireless application protocol?

10. Write a HTML program for a single web page.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain about the signal propagation in detail.

Or

(b) Describe about the antennas.

12. (a) Describe about the digital audio broadcasting.

Or

(b) Explain the DDMA technique.

13. (a) Describe the IEEE 802.11 protocol in detail.

Or

(b) Describe about the wireless ATM working group.

14. (a) Explain about the traditional TCP 292.

Or

(b) Explain the functions of dynamic host configuration protocol.

15. (a) Explain the HTTP protocol.

Or

(b) What are the features of HTML language?

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe about the wireless transmission.

17. Describe about the TETRA, UMTS systems.

18. Describe about the wireless ATM.

19. Describe the transaction oriented tap.

20. Describe the approaches that might help wireless access.

———————

AFC–10396

Sub. Code
1BCEE3B

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Computer Science

Elective – VISUAL BASIC

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the two types of a string variable?

2. What is static array?

3. What is a Combo box control? What is its use?

4. What are the properties of menus?

5. What is the difference between image and picture box control?

6. Write the syntax of line method.

7. Why we use Treeview control?

8. What are the different types of dialog box?

9. What are two validate with data control?

10. What is ADO? What are its objects?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain nested control structures in VB.

Or

(b) Discuss about collections.

12. (a) Discuss the appearance of forms in detail.

Or

(b) Explain the basic properties of List Box and Combo Box.

13. (a) Explain loading and saving images.

Or

(b) Discuss about scale properties and methods.

14. (a) Explain the properties of common dialogs control.

Or

(b) Explain the properties of RichTexBox control.

15. (a) Explain the structure of BIBLIO database.

Or

(b) Discuss about DataEnvironment with data Grid control and MSHFlexGrid control.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain loop statements with examples.

17. Discuss in detail about the scroll bar and slider controls.

18. Explain any three graphics method with examples.

19. Discuss in detail about multiple document interface.

20. Explain the basic properties and methods of data control.

AFC–10497

Sub. Code
1BCESA1/ 1BCEA1

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

Allied — DIGITAL ELECTRONICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the informations about ASCII code.

2. What is meant by Excess – 3 code?

3. Write the technical configurations of multiplexers.

4. What is meant by PLA?

5. Write the truth table of RS-flipflop.

6. What are the types of shift registers?

7. Write the principle of MOS technology.

8. Write the principle of CMOS technology.

9. Draw the circuit of resistive divider.

10. Draw the circuit of 4-bit D/A converter.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Prove that $A(A' + C)(A'B + C)(A'BC + C') = 0$.

Or

(b) Simplify the following Boolean equation.

$y = A'B'C' + A'BC' + AB'C' + ABC'$.

12. (a) Explain about the magnitude comparator.

Or

(b) Describe about the 1-OF-16 decoder.

13. (a) Explain with function table of JK flipflop.

Or

(b) Explain about ring counters.

14. (a) Explain about DTL circuits.

Or

(b) Describe the bipolar transistor characteristics.

15. (a) Describe the dual slope A/D conversion.

Or

(b) Explain about the binary ladder.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain about the positive and negative logic.

17. Describe the binary adder – substractor.

18. Explain about the clocked RS – flipflops.

19. Describe the TTL logic.

20. Describe the continuous A/D conversion techniques.

———————

AFC–10498

Sub. Code
1BCESA2

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

Allied – MICROPROCESSORS AND INTERFACING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the functions of pointers and index registers of 8086 microprocessor.

2. Write the functions of direction flag of 8086 microprocessor.

3. Write the syntax and function of register indirect addressing mode.

4. What are the functions of stack in 8088 microprocessor.

5. Draw the pin diagram of 8253 PIT.

6. What are the operating modes of 8253?

7. Draw the internal architecture of 8272 Floppy Disk controller.

8. Write the Read/Write control logic of 8275 CRT controller.

9. What are the flag registers available in 80286 processor.

10. What are the data types supported by 80386 processor.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the flag registers of 8086 microprocessor.

Or

(b) Explain about the memory timings of 8086 processor.

12. (a) Write a assembly language program using 8086/8088 microprocessor instruction set for addition of two numbers.

Or

(b) Describe about sub routines.

13. (a) Explain about 8255 programmable interrupt controller.

Or

(b) Explain about 8288 bus controller.

14. (a) Explain how to interface modern with the 8088 microprocessor.

Or

(b) Describe about the signal descriptions of 8275 CRT controller.

15. (a) Explain the interrupts of 80286 microprocessor.

Or

(b) Explain about the read address mode of 80386 microprocessor.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain about the address bus, data bus, control bus of 8088 microprocessor.

17. Describe about the assembler directives.

18. Describe about the 74LS245 transceiver of micro computer system.

19. Describe about the system ROM.

20. Explain the register organization of 80386 micro processor.

AFC–10499

Sub. Code
1BCESA3

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

Allied : RESOURCE MANAGEMENT TECHNIQUES

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the role of operations research in decision making?

2. Define a scientific model.

3. Differentiate canonical and standard forms of linear programming problem.

4. Define feasible solution of a LPP.

5. What is duality? What is the significance of dual variable in simplex solution?

6. Write the mathematical form of IPP.

7. Define assignment problem.

8. Write any two applications of IPP.

9. Define degenerate basic fesible solution of TPP.

10. What are balanced and unbalanced transportation problems?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the development of operations research.

Or

(b) Briefly explain the scope of OR.

12. (a) Solve graphically.

Minimize $z = 5x_1 + 2x_2$

Subject to :

$3x_1 + x_2 \geq 3$

$3x_1 - 2x_2 \leq 6$

$x_1 + x_2 \leq 4$

$x_1, x_2 \geq 0.$

Or

(b) Solve the following LPP using simplex method.

Maximize $z = 4x_1 + 10x_2$

Subject to :

$2x_1 + x_2 \leq 50$

$2x_1 + 5x_2 \leq 100$

$2x_1 + 3x_2 \leq 90$

$x_1, x_2 \geq 0.$

13. (a) Using duality solve.

Maximize $z = 4x_1 + 3x_2$

Subject to :

$2x_1 + x_2 \leq 10$

$x_1 + x_2 \leq 8$

$x_1, x_2 \geq 0.$

Or

(b) Explain the procedure to solve the LPP using dual simplex method.

14. (a) Explain the procedure to solve an assignment problem.

Or

(b) Solve the following assignment problem.

	I	II	III	IV	V
A	8	4	2	6	1
B	0	9	5	5	4
C	3	8	9	2	6
D	4	3	1	0	3
E	9	5	8	9	5

15. (a) Explain North-West corner rule to obtain an initial basic feasible solution for the transportation problem.

Or

(b) Obtain an initial basic feasible solution using the row minima method.

	A	B	C	
I	50	30	220	1
II	90	45	170	3
III	250	200	50	4
	4	2	2	

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the features of OR.

17. Solve using two-phase simplex method.

Maximize $z = 5x_1 + 8x_2$

Subject to :

$3x_1 + 2x_2 \geq 3$

$x_1 + 4x_2 \geq 4$

$x_1, x_2 \leq 5$

$x_1, x_2 \geq 0.$

18. Using the dual simplex method solve.

Minimize $z = x_1 + 2x_2 + 3x_3$

Subject to :

$x_1 - x_2 + x_3 \geq 4$

$x_1 + x_2 + 2x_3 \leq 8$

$x_2 - x_3 \geq 2$

$x_1, x_1, x_3 \geq 0.$

19. Solve the following travelling salesman problem.

	A	B	C	D
A	∞	4	7	3
B	4	∞	6	3
C	7	6	∞	7
D	3	3	7	∞

20. Solve the following transportation problem.

	D_1	D_2	D_3	D_4	
S_1	6	8	8	5	30
S_2	5	11	9	7	40
S_3	8	9	7	13	50
	35	28	32	25	

AFC–10500

Sub. Code
1BCESA4

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

Allied — COMPUTER ORIENTED NUMERICAL METHODS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the condition for convergence in iteration method?

2. Differentiate Gauss-Jacobi and Gauss-Seidel iteration methods.

3. Define the principle of least squares.

4. Define eigen vectors.

5. Define forward and backward difference operator.

6. Write the formula for inverse interpolation.

7. What is the first derivative and second derivative of Newton's forward formula?

8. Write Gaussian quadrative formula.

9. Define Taylor's series method.

10. Write Runge Kutta fourth order formula.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Find a positive root of the following equation by bisection method.

$x^3 - 4x - 9 = 0$.

Or

(b) Explain Regula Falsi method.

12. (a) Explain methods of moments for cure fitting.

Or

(b) Find the eigen values and eigen vector of the matrix

$$\begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$$

13. (a) Find $y(-1)$ if $y(0) = 2$, $y(1) = 9$, $y(2) = 28$, $y(3) = 65$, $y(4) = 126$, $y(5) = 217$

Or

(b) Derive Newton's backward interpolation formula.

14. (a) Evaluate $\int_1^2 \frac{dx}{x}$ using Simpson's rule take $h = 0.25$.

Or

(b) Explain Romberg's method.

15. (a) Using Taylor series method, find, correct to four decimal places, the value of $y(0.1)$, given $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$.

Or

(b) Explain Euler's method.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Solve the following systems by Gauss-elimination method

$3x + 4y + 5z = 18$
$2x - y + 8z = 13$
$5x - 2y + 7z = 20.$

17. Fit a straight line to the data given below :

x :	5	10	15	20	25
y :	15	19	23	26	30

18. Using Lagrange's interpolation formula, find $y(10)$ from the following table.

x :	5	6	9	11
y :	12	13	14	16

19. Find the first and second derivative of the function tabulated below at $x = 0.6$.

x :	0.4	0.5	0.6	0.7	0.8
y :	0.5836	1.7974	2.0442	2.3275	2.6511

20. Given $\frac{dy}{dx} = \frac{1}{2}(x + y)$ and $y(0) = 2$, $y(0.5) = 2.636$, $y(1) = 3.593$ and $y(1.5) = 4.968$ Evaluate $y(2)$ by Milne's predictor corrector method.

AFC–10502

Sub. Code
1BCEA3

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

Allied — PROGRAMMING IN C

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A (10 × 1.5 = 15)

Answer **all** questions.

1. What is function oriented programming language?

2. What is constant? Give an example.

3. How do declare one dimensional array?

4. How do initialize string variable?

5. What is recursion?

6. What is union?

7. What are the uses of * operator in pointers?

8. What is pointer?

9. What is sequential file?

10. Define macro.

Part B (5 × 3 = 15)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain Arithematic and relational operators in 'C'.

Or

(b) Explain switch case statement with an example.

12. (a) Explain arithmetic operations on characters.

Or

(b) Explain any three string functions.

13. (a) Explain array at structures with an example.

Or

(b) Explain structure with in structure with an example.

14. (a) Explain pointers and functions.

Or

(b) Explain printers and arrays.

15. (a) Explain about command line arguments.

Or

(b) Write short notes on error handling in files.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain data type supported by C.

17. Write a 'C' program to sort ten numbers.

18. What is user defined function? Explain how are parameters passed into the function.

19. Explain pointers and structures with suitable example.

20. Explain sequential file operations.

———————

AFC–10503

Sub. Code
1BCEA4

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Computer Science

Allied – PROGRAMMING IN C++

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 60 Marks

Part A (10 × 1.5 = 15)

Answer **all** questions.

1. Define Data abstraction.

2. What is Exit Control loop?

3. Define an Object.

4. What is static data member?

5. Define Pure virtual function.

6. How do display formatted output in C++?

7. What is Random Access file?

8. What is function Template?

9. What is meant by Exception?

10. How do throw an exception?

Part B (5 × 3 = 15)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain rules for coding for loops.

Or

(b) Compare switch case and if....else if statement.

12. (a) Explain multiple constructor in a class.

Or

(b) Explain friend function.

13. (a) Explain C++ output stream class.

Or

(b) Explain Polymorphism.

14. (a) Explain Generic function.

Or

(b) Explain Class Template.

15. (a) How do handle exception in constructor?

Or

(b) How do handle exception in inheritance?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss features of Object Oriented Programming.

17. Explain passing objects to a function as argument.

18. Explain multiple and Hierarchical inheritance.

19. Explain any five file operations.

20. Discuss Exception Handling model.